

Replication Assignment: Minimum Wages and Employment

Card & Krueger (1994)

Purpose

This optional replication assignment is part of the mentorship component of the RA lab. Its goal is to help you develop *transferable skills* used in applied economic research: clean data construction, visualization, and causal reasoning.

This assignment is intentionally lightweight. It is **not** meant to take substantial time away from your primary RA responsibilities.

Background

In April 1992, New Jersey increased its minimum wage from \$4.25 to \$5.05 per hour. Pennsylvania did not. Card & Krueger (1994) study employment at fast-food restaurants in both states before and after the policy change.

The paper uses a *difference-in-differences* (DiD) design to estimate the causal effect of the minimum wage increase on employment.

Software & Workflow

You will use:

- **R** (data cleaning, visualization, estimation)
- **Git & GitHub** (version control)

You will create a GitHub repository and add me as a collaborator.

Data

You may obtain the data from one of the following sources:

- Card's replication files:
https://davidcard.berkeley.edu/data_sets.html
- An R-friendly version via Rdatasets (search for Card–Krueger minimum wage data):
<https://vincentarelbundock.github.io/Rdatasets/>

Clearly document which source you use.

Part 0: Framing the Causal Question (Conceptual)

Answer the following in short bullet points:

1. What is the causal question?
2. What is the treatment?
3. What defines the treated and control groups?
4. What is the outcome variable?
5. Why would a simple post-policy comparison be misleading?

Part 1: Data Construction

Starting from the raw data:

1. Clean variable names.
2. Construct:
 - a treatment indicator (New Jersey),
 - a post-policy indicator,
 - their interaction.
3. Save a cleaned dataset.

Your goal is to produce a dataset you would feel comfortable handing to a coauthor.

Part 2: Visualization

Before running regressions, create at least two figures:

1. Average employment over time by treatment status.
2. A figure that helps assess whether pre-treatment trends look similar.

All figures should be clearly labeled and saved to your repository.

Part 3: Difference-in-Differences Estimation

Estimate:

$$Y_{it} = \alpha + \beta \text{Treated}_i + \gamma \text{Post}_t + \delta(\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}.$$

- Report the estimate of δ .
- Interpret it in words.
- Compare it loosely to the main result in the paper.

Part 4: Identification

In one short paragraph:

- State the key identifying assumption.
- Give one reason it might fail.
- Comment on whether your plots support its plausibility.

Final Deliverables

- A GitHub repository with clean, commented code
- Two figures
- A **one-page memo (maximum)** summarizing your results and interpretation

Helpful References

- Cunningham, *Causal Inference: The Mixtape* (DiD chapter):
<https://mixtape.scunning.com/difference-in-differences.html>
- Wickham & Grolemund, *R for Data Science*:
<https://r4ds.had.co.nz/>
- Happy Git with R:
<https://happygitwithr.com/>